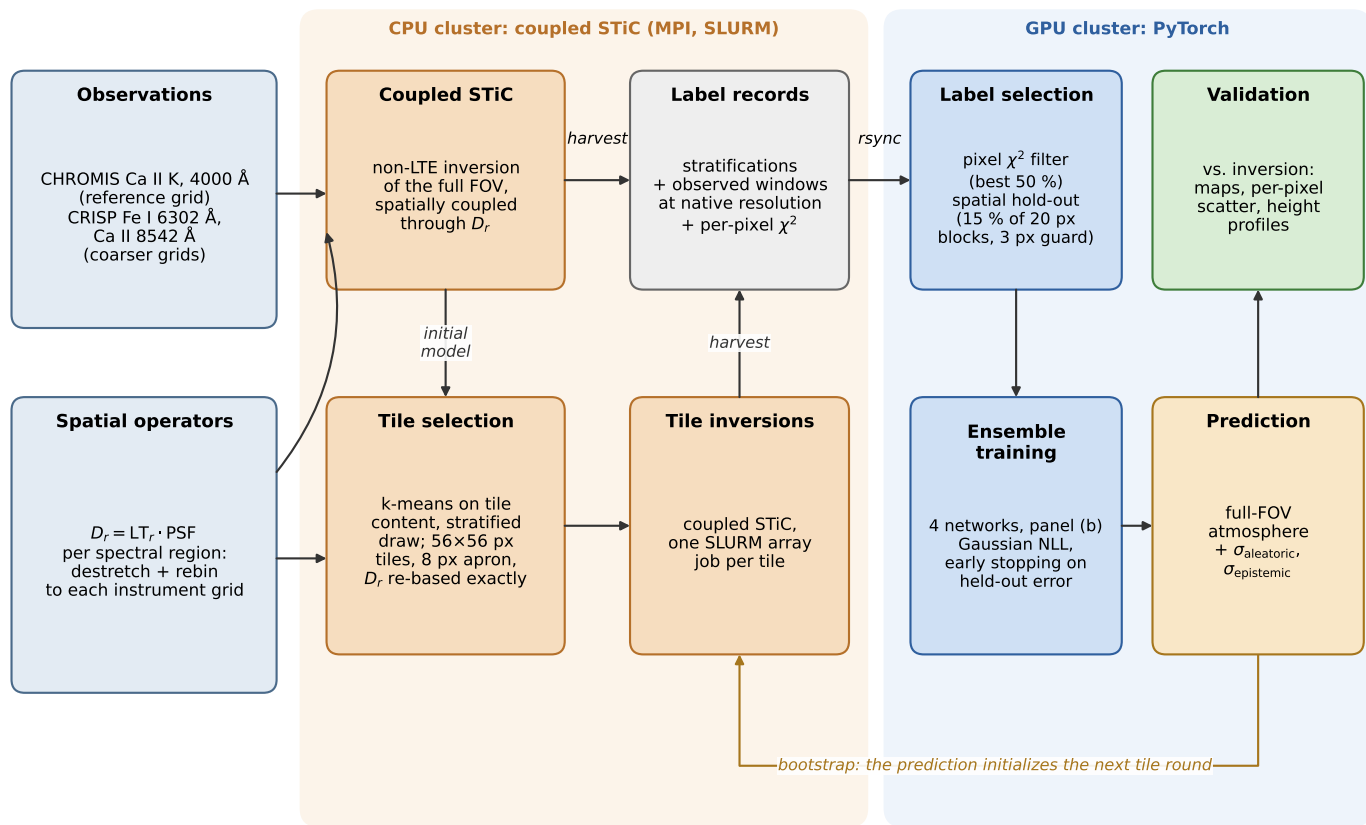
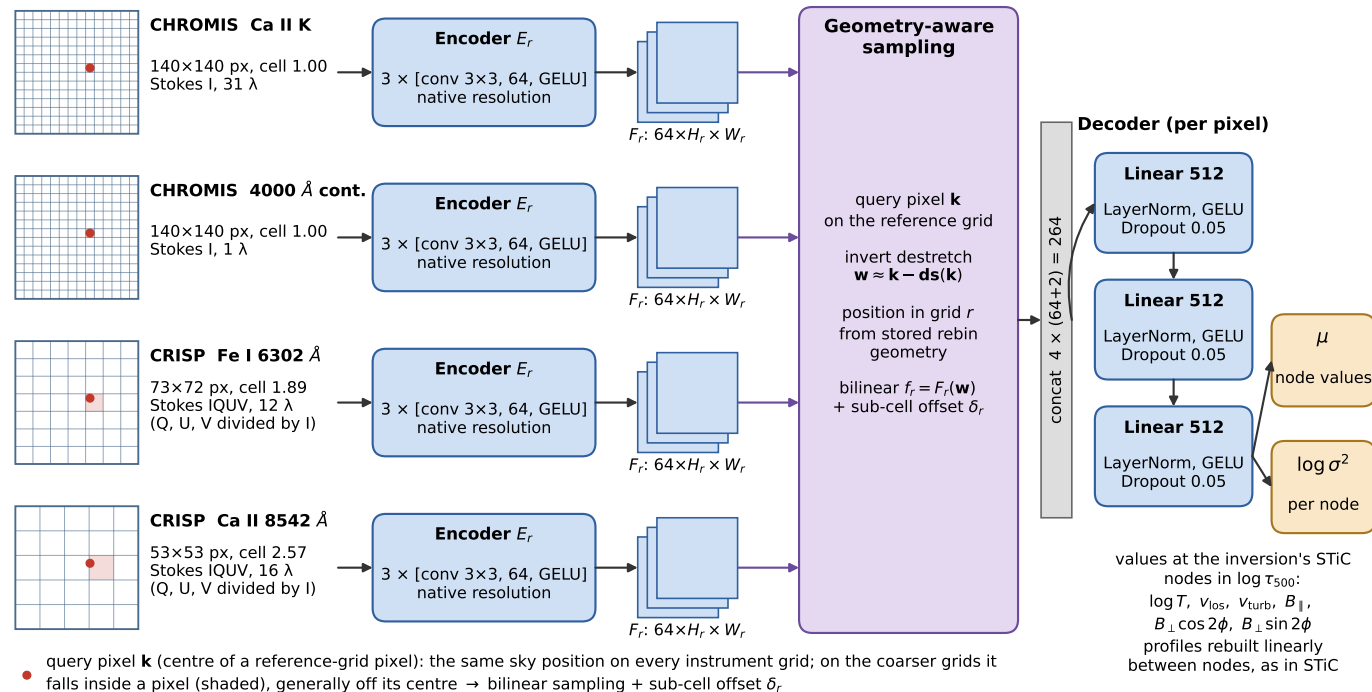


(a) Label generation, training and the bootstrap loop



(b) Multi-resolution emulator network (one ensemble member)



Training and uncertainty

Loss: Gaussian negative log-likelihood $\frac{1}{2} [\log \sigma^2 + (y - \mu)^2 / \sigma^2]$ on standardized targets, plus $\lambda \sum (\partial^2 \mu / \partial \log \tau^2)^2$ with $\lambda = 10^{-3}$

AdamW (lr 3×10^{-4} , cosine schedule), $\leq 20\,000$ steps of 4 records $\times \leq 2048$ pixels, early stopping on the held-out prediction error

Ensemble of 4 members (different initializations): prediction = ensemble mean, $\sigma_{\text{aleatoric}} = (\sigma^2)^{1/2}$, $\sigma_{\text{epistemic}}$ = spread of the member means